

Divita Saraogi

Data Scientist / AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Results-driven Data Scientist with 5+ years of experience in Machine Learning, AI, Predictive Analytics, and Statistical Modeling across scientific and industrial domains. Currently working at Picarro Technologies on AI-driven gas spectral analysis and compound fingerprinting using Python, PyTorch, ML models, Deep learning and AWS. Skilled in building scalable ML pipelines, analyzing complex datasets, and delivering data-driven solutions through strong industry and PhD-level research expertise.

TECHNICAL SKILLS

Programming & Data Analysis:	Python, SQL, R, Pandas, NumPy, Excel
Machine Learning & AI:	PyTorch, TensorFlow, Scikit-Learn, Deep Learning, Predictive Modeling, Feature Engineering
Data Visualization & BI:	Power BI, Tableau, Matplotlib, Seaborn
Cloud, Tools & Development:	AWS, Git, GitHub, Docker, PyCharm, Jupyter Notebook
Scientific & Research Computing:	Spectral Analysis, Scientific Data Processing, Statistical Modeling, Experimental Data Analysis

PROFESSIONAL EXPERIENCE

Data Scientist – AI Team | Picarro Technologies India Pvt. Ltd.

Jun 2025 – Present | Bengaluru, India

- Working on AI-driven analysis of gas spectral data for compound speciation and quantification.
- Developed fingerprinting methodologies to identify and speciate compounds from spectral datasets.
- Built ML and deep learning pipelines for analytical workflows.
- Worked with AWS-based infrastructure for scalable data handling and experimentation.
- Collaborated with cross-functional teams to improve model accuracy and analytical reliability.
- Performed feature extraction, preprocessing, and optimization for high-dimensional spectral data.

Data Scientist | TossCall Services India Pvt. Ltd.

Feb 2024 – Feb 2025 | Bhilai, India

- Developed Power BI dashboards to visualize customer behavior and business KPIs.
- Built ML models for customer return prediction and business analytics.
- Conducted exploratory data analysis using Python, Matplotlib, and Seaborn.
- Optimized SQL queries for large-scale datasets and reporting efficiency.
- Collaborated with stakeholders to provide data-driven recommendations.

Research Scholar (PhD) | IIT Bombay

Jan 2018 – Jan 2024 | Mumbai, India

- Developed real-time Gamma-Ray Burst localization pipelines using AstroSat data.
 - Performed statistical analysis and spectral modeling on large scientific datasets.
 - Built novel localization methodologies using probabilistic and signal-processing techniques.
 - Published peer-reviewed research papers and presented at international conferences.
 - Worked extensively with scientific computing, Bayesian analysis, and data interpretation.
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EDUCATION

Ph.D. in Physics (2024)

Indian Institute of Technology Bombay, Mumbai

M.Sc. in Physics (2014)

Sant Gadge Baba Amravati University, Amravati

B.Sc. in Physics (2012)

LAD College, Nagpur

PROJECT HIGHLIGHTS

- AI-Based Gas Spectral Fingerprinting – Developed fingerprinting solutions for gas compound identification and speciation.
 - Customer Return Prediction – Built ML models to predict customer return behavior and optimize business insights.
 - Real-Time GRB Localization Pipeline – Developed automated scientific pipelines for Gamma-Ray Burst localization using AstroSat data.
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CERTIFICATIONS

- Microsoft SQL
 - Python for Data Science
 - Power BI Certification
 - Statistics & Mathematics for Data Science
 - Advanced Excel
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AWARDS & ACHIEVEMENTS

- ASI Best Poster Award (2024) for outstanding research presentation on Gamma-Ray Burst localization.
- Published multiple peer-reviewed research papers in astrophysics and scientific data analysis.
- Delivered research presentations at scientific conferences and outreach programs.